Subgraph retrieval quality

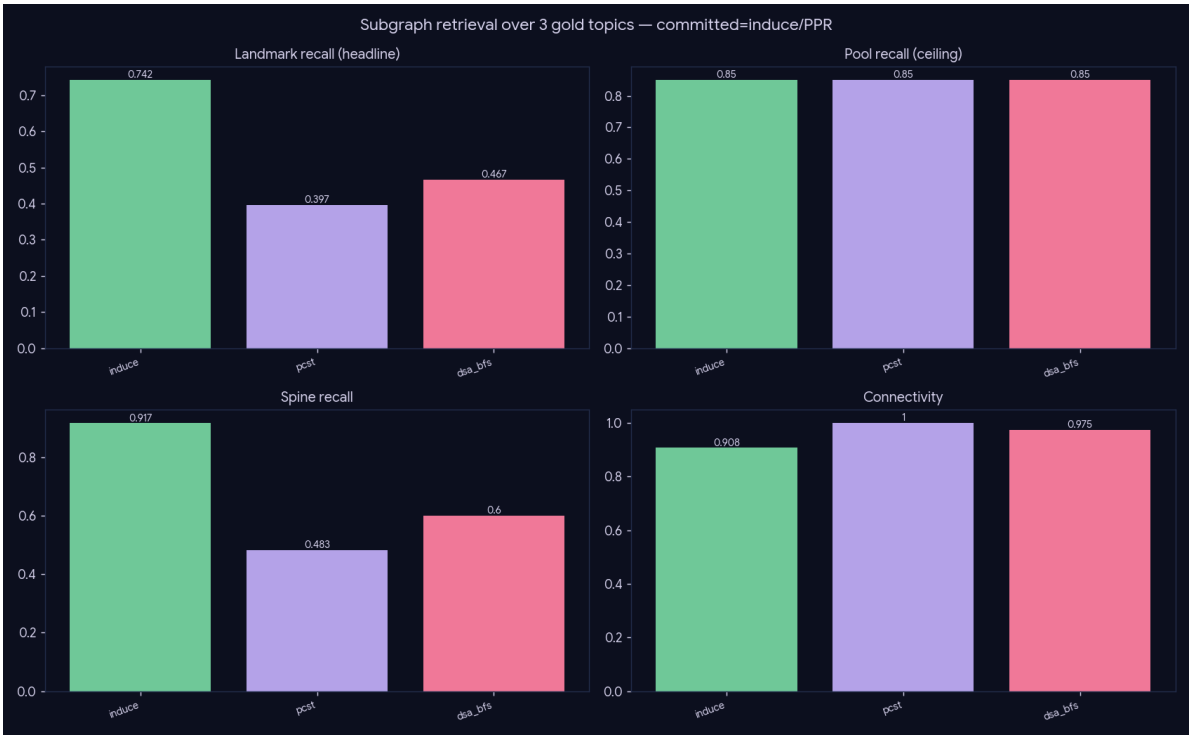
arxivkg · retrieval · the committed recipe in isolation

What we did

The committed retrieval recipe — hybrid pool (asym cosine ∪ title-BM25) → backward-ancestor augmentation → induce citation subgraph → Personalized-PageRank landmark ranking — is locked; this measures the one thing it does: recovering a field's *canonical landmark papers* as a bounded, connected, on-topic citation subgraph. **Retrieval level only: no LLM, no compose, no judge.** Over each gold topic we build one shared candidate pool and run three Stage-4 selectors on it: `induce` (the committed PPR), `pcst` (Prize-Collecting Steiner Tree, G-Retriever 2402.07630), and `dsa_bfs` (similarity-aware BFS, DynaGRAG 2412.18644). Splitting `pool_recall` (the reachability ceiling) from `landmark_recall` (after ranking) localizes any loss: a gap is a *ranking* problem, a low ceiling a *reach* problem.

Results (3 gold topics: diffusion, transformer, testtime)

method	pool rec.	landmark rec.	spine rec.	conn.	cosine
induce (committed)	0.85	0.742	0.917	0.908	0.497
pcst	0.85	0.397	0.483	1.0	0.602
dsa_bfs	0.85	0.467	0.6	0.975	0.577



Conclusion

The committed `induce`/PPR method recovers 0.742 of canonical landmarks and 0.917 of the foundational spine, against a shared pool ceiling of 0.85 — so PPR ranking retains $0.742/0.85 \approx 87\%$ of everything reachable, and almost the entire origin story. The 0.85 ceiling (not 1.0) is a pool-*reach* limit, not a ranking flaw: the missing canonical papers fall outside the topical cosine cloud or are uncited within it — a candidate-pool question (wider `n_pool` / more ancestor hops), not a selector one. `induce` **dominates every recall axis.** `pcst` wins *only* connectivity (1.0) and mean-cosine while *halving* landmark recall (0.40) — its cosine prize $\neq$ citation centrality, so it keeps tight high-cosine clusters and drops central-but-not-top-cosine canonical papers. Membership must be decided by PPR; connectivity is a downstream *presentation* concern (the lineage tree adds Steiner bridges *over* the PPR landmarks, buying PCST-level connectivity without its recall cost). **Caveat:** n= 3 (the only topics with curated gold); per-topic numbers are stable (`induce` landmark_recall = 0.625/0.60/1.0), but a wider gold battery is the cheapest way to harden the claim.